3(a)	point where (all) the weight (of an object) is taken to act	**B1**
3(b)(i)	vertical component $= 45 \sin 37°$ $= 27$ N	**A1**
3(b)(ii)	the magnitudes of the three moments about A are (23×0.48), (27×0.56) and $(W \times 0.76)$ correct magnitude of any one moment about A	**C1**
	correct magnitudes of any two moments about A	**C1**
	$(23 \times 0.48) + (W \times 0.76) = 27 \times 0.56$ $W = 5.4$ N	**A1**
3(b)(iii)	horizontal component $= 45 \cos 37°$ $= 36$ N	**A1**
3(b)(iv)	decrease	**B1**
3(b)(v)	$\sigma = F / A$	**C1**
	$\sigma = F / \pi r^2$ or $4F / \pi d^2$ so $\sigma = 5.3 \times 10^7 \times \pi r^2 / \pi(3r)^2$ $= 5.3 \times 10^7 / 9$ $= 5.9 \times 10^6$ Pa	**A1**